

MS&E 228: Applied Causal Inference Powered by ML and AI

Lecture 15: Instrumental Variables

Vasilis Syrgkanis

Stanford University

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Goals of this Lecture

What can we do when we have unobserved confounding?

1. **The IV Idea:** How can we use a special variable (an instrument) to isolate causal effects?
2. **The LATE:** What is the Local Average Treatment Effect and who are the *compliers*?
3. **Estimation:** How do we estimate IV models using Doubly Robust and Double ML methods?
4. **Weak Instruments:** How can we perform robust inference when the instrument is weak?

Motivating Example: The Eager Trial

We return to the Eager trial, which examined the effect of low-dose aspirin (D) on pregnancy (Y).

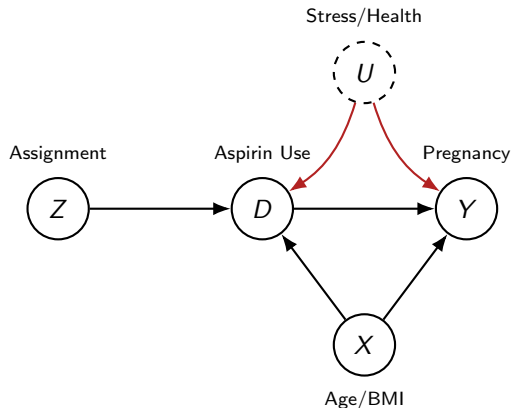
- ▶ Patients were randomly assigned to a treatment group ($Z = 1$ for aspirin, $Z = 0$ for placebo).
- ▶ However, some patients did not adhere to their assignment (e.g., forgot to take the pills).
- ▶ This is a problem of **non-compliance**.

The Confounding Problem

The reasons for non-compliance could be driven by unobserved factors (U) like stress or health habits, which might also affect the pregnancy outcome (Y). This introduces confounding, even though the initial assignment (Z) was random.

The Instrumental Variable (IV) Causal Graph

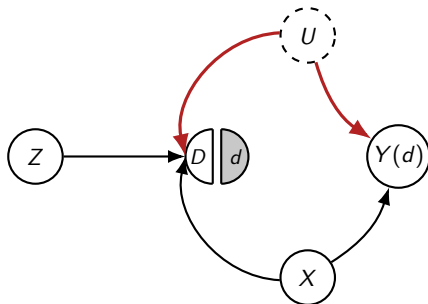
This scenario is described by the canonical IV graph.



Key Idea: The assignment Z is called an **instrument**. It affects the outcome Y only through its effect on the treatment D .

Checking Conditional Exogeneity with the SWIG

Let's draw the SWIG for the intervention $\text{fix}(D = d)$ and check for conditional exogeneity.



Backdoor Path is Open

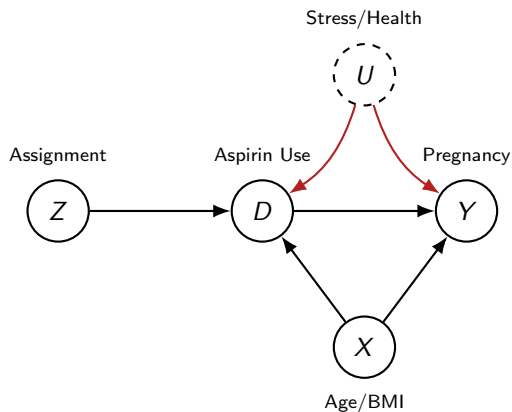
The path $D \leftarrow U \rightarrow Y(d)$ is open because U is unobserved. Conditional exogeneity $Y(d) \perp\!\!\!\perp D \mid X$ **fails**. We cannot use our standard DR/DML estimators.

Part II: Identification with Instrumental Variables

From a Naive Ratio to the LATE

Can we Express the Target Effect as a Function of Identifiable Effects

What effects are identifiable in this Causal Graph?



A Simplistic (but Wrong) Approach

Since the effect of Z on Y is fully mediated by D , can we decompose the effects?

Idea: Maybe the total effect is the product of the intermediate effects.

$$\text{ATE}(Z \rightarrow Y) = \text{ATE}(Z \rightarrow D) \times \text{ATE}(D \rightarrow Y)$$

If this were true, we could solve for the effect we want:

$$\text{ATE}(D \rightarrow Y) = \frac{\text{ATE}(Z \rightarrow Y)}{\text{ATE}(Z \rightarrow D)} = \frac{\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]}{\mathbb{E}[D|Z=1] - \mathbb{E}[D|Z=0]}$$

The numerator and denominator are identified because Z is random (unconditionally exogenous). But is the formula correct?

Decomposition at the Individual Effect Level

The correct decomposition is at the individual level:

$$\text{ITE}(Z \rightarrow Y) = \text{ITE}(Z \rightarrow D) \times \text{ITE}(D \rightarrow Y)$$

Always true, by the fact that Z, D are binary and effect of Z is fully mediated by D :

$$\begin{aligned} \underbrace{Y(Z=1) - Y(Z=0)}_{\text{ITE}(Z \rightarrow Y)} &= \underbrace{Y(D(Z=1)) - Y(D(Z=0))}_{\text{Effect of } Z \text{ fully mediated by } D} \\ &= \underbrace{(D(Z=1) - D(Z=0))}_{\text{ITE}(Z \rightarrow D)} \underbrace{(Y(D=1) - Y(D=0))}_{\text{ITE}(D \rightarrow Y)} \end{aligned}$$

The Catch: Heterogeneity

The correct decomposition is at the individual level:

$$\text{ITE}(Z \rightarrow Y) = \text{ITE}(Z \rightarrow D) \times \text{ITE}(D \rightarrow Y)$$

Taking expectations, we get:

$$\text{ATE}(Z \rightarrow Y) = \mathbb{E}[\text{ITE}(Z \rightarrow D) \times \text{ITE}(D \rightarrow Y)]$$

The Problem

In general, the expectation of a product is not the product of expectations:

$$\mathbb{E}[A \cdot B] \neq \mathbb{E}[A] \cdot \mathbb{E}[B]$$

This equality only holds if A and B are uncorrelated.

The Role of Homogeneity

The naive formula would work if we assumed the individual effects were uncorrelated:

$$\text{ITE}(Z \rightarrow D) \perp\!\!\!\perp \text{ITE}(D \rightarrow Y)$$

- ▶ $\text{ITE}(Z \rightarrow D)$ is the **compliance effect**: does the individual comply with the assignment?
- ▶ $\text{ITE}(D \rightarrow Y)$ is the **treatment effect**: how does the individual respond to the treatment?

This assumption means that "*who complies*" is uncorrelated with "*who responds to treatment*". This is a strong assumption that may not hold in many real-world settings.

One Idea: Relax Assumption by Performing Conditional Analysis

If individual effects were uncorrelated conditional on observables X :

$$\text{ITE}(Z \rightarrow D) \perp\!\!\!\perp \text{ITE}(D \rightarrow Y) \mid X$$

Then we can break the expectation of the product:

$$\text{CATE}(Z \rightarrow Y \mid X) = \text{CATE}(Z \rightarrow D \mid X) \times \text{CATE}(D \rightarrow Y \mid X)$$

Then we can identify the ATE as the expectation of the ratio of CATEs:

$$\text{ATE}(D \rightarrow Y) = \mathbb{E}[\text{CATE}(D \rightarrow Y \mid X)] = \mathbb{E} \left[\frac{\text{CATE}(Z \rightarrow Y \mid X)}{\text{CATE}(Z \rightarrow D \mid X)} \right]$$

Both CATEs in the ratio can be identified by conditional exogeneity. For $O \in \{D, Y\}$

$$\text{CATE}(Z \rightarrow O \mid X) = \mathbb{E}[\mathbb{E}[O \mid Z = 1, X] - \mathbb{E}[O \mid Z = 0, X]]$$

A New Idea: Re-interpreting the Ratio

Instead of making the strong (conditional) homogeneity assumption, let's change our **interpretation** of what the IV ratio identifies.

This leads to one of the most important concepts in modern econometrics: the **Local Average Treatment Effect (LATE)**.

To define it, we need two key assumptions:

1. **Exclusion Restriction:** The instrument Z affects the outcome Y only through the treatment D . (This is encoded in our DAG).
2. **Monotonicity:** The instrument can only encourage, not discourage, treatment uptake.

The "No Defiers" (Monotonicity) Assumption

Definition: Monotonicity

For every individual, being assigned to the treatment group makes them weakly more likely to take the treatment:

$$D_i(Z = 1) \geq D_i(Z = 0) \quad \forall i$$

This assumption rules out the existence of **defiers**—individuals who would always do the opposite of what they are assigned.

Compliers, Always-Takers, and Never-Takers

Under monotonicity, the population can be divided into three groups based on their potential treatment behavior:

Group	Potential Behavior	ITE($Z \rightarrow D$)
Compliers	$D(1) = 1, D(0) = 0$	1
Always-Takers	$D(1) = 1, D(0) = 1$	0
Never-Takers	$D(1) = 0, D(0) = 0$	0

Key Insight: The instrument only changes the behavior of the **compliers**.

LATE as the ATE for Compliers

Revisit the ATE decomposition. Since $\text{ITE}(Z \rightarrow D)$ is 1 for compliers and 0 for others:

$$\begin{aligned}\text{ATE}(Z \rightarrow Y) &= \mathbb{E}[\text{ITE}(D \rightarrow Y) \cdot \text{ITE}(Z \rightarrow D)] = \mathbb{E}[\text{ITE}(D \rightarrow Y) \cdot \mathbf{1}\{\text{Complier}\}] \\ &= \mathbb{E}[\text{ITE}(D \rightarrow Y) | \text{Complier}] \cdot \Pr(\text{Complier})\end{aligned}$$

Moreover, $\Pr(\text{Complier}) = \mathbb{E}[\mathbf{1}\{\text{Complier}\}] = \mathbb{E}[\text{ITE}(Z \rightarrow D)] = \text{ATE}(Z \rightarrow D)$.

The LATE Result

$$\frac{\text{ATE}(Z \rightarrow Y)}{\text{ATE}(Z \rightarrow D)} = \mathbb{E}[\text{ITE}(D \rightarrow Y) | \text{Complier}] = \mathbf{LATE}$$

The IV ratio identifies the ATE for the subpopulation of **Compliers**.

Poll Question

Poll: Understanding LATE

In the Eager trial, the LATE represents the average treatment effect of aspirin for which group?

1. All women in the study.
2. Women who would take aspirin if assigned to, and not take it if assigned to placebo.
3. Women who would take aspirin regardless of their assignment.
4. Women who would never take aspirin regardless of their assignment.

Part III: Estimation and Inference

From Identification to Estimation

Estimation of LATE

When the instrument Z is unconditionally random, the LATE is identified by the ratio of two simple difference-in-means estimators:

$$\widehat{\text{LATE}} = \frac{\mathbb{E}_n[Y|Z=1] - \mathbb{E}_n[Y|Z=0]}{\mathbb{E}_n[D|Z=1] - \mathbb{E}_n[D|Z=0]}$$

More generally, if the instrument is only conditionally random given covariates X , i.e., $Z \perp\!\!\!\perp \{Y(z), D(z)\} \mid X$, we need to adjust for X .

$$\text{LATE} = \frac{\mathbb{E}[\mathbb{E}[Y|Z=1, X] - \mathbb{E}[Y|Z=0, X]]}{\mathbb{E}[\mathbb{E}[D|Z=1, X] - \mathbb{E}[D|Z=0, X]]}$$

This is a ratio of two g-formula estimators.

Doubly Robust Estimation of LATE

To gain robustness, we use doubly robust estimators for the numerator and denominator.

DR-LATE Estimator

$$\widehat{\text{LATE}} = \frac{\widehat{\text{DR}}(Z \rightarrow Y)}{\widehat{\text{DR}}(Z \rightarrow D)}$$

where, for an outcome $O \in \{Y, D\}$:

$$\widehat{\text{DR}}(Z \rightarrow O) = \frac{1}{n} \sum_{i=1}^n \left[\hat{\mu}_O(1, X_i) - \hat{\mu}_O(0, X_i) + \frac{Z_i - \hat{p}(X_i)}{\hat{p}(X_i)(1 - \hat{p}(X_i))} (O_i - \hat{\mu}_O(Z_i, X_i)) \right]$$

$\hat{\mu}_O(z, x) = \mathbb{E}[O|Z=z, X=x]$ is the outcome regression; $\hat{p}(x) = \mathbb{E}[Z|X=x]$ is the propensity score.

Asymptotic Normality of DR-LATE

Because the DR estimators are insensitive to first-stage estimation error, we can derive the asymptotic distribution of the LATE estimator.

$$\sqrt{n}(\widehat{\text{LATE}} - \text{LATE}) \xrightarrow{d} N(0, \text{Var}(\rho(W)))$$

where $W = (X, Z, D, Y)$ and the influence function $\rho(W)$ is given by:

$$\rho(W) = \frac{\phi_Y(W) - \text{LATE} \cdot \phi_D(W)}{\mathbb{E}[\phi_D(W)]}$$

and $\phi_O(W)$ is the influence function for the DR estimator of the effect of Z on O .

This allows us to construct valid confidence intervals for the LATE.

Python Pseudocode for DR-LATE

```
def dr_late(X, Z, D, y, models, nfolds=5, trim=0.01):  
    # Cross-fit nuisance:  $E[Y|X, Z=z]$ ,  $E[D|X, Z=z]$ ,  $E[Z|X]$   
    yhat0, yhat1 = cross_fit_predict(models['y'], X, y, Z)  
    dhat0, dhat1 = cross_fit_predict(models['d'], X, D, Z)  
    zhat = cross_val_predict(models['z'], X, Z, method='predict_proba')[:,1]  
    zhat = np.clip(zhat, trim, 1 - trim)  
    # Observed nuisance values  
    yhat = yhat0 * (1-Z) + yhat1 * Z  
    dhat = dhat0 * (1-Z) + dhat1 * Z  
    # Doubly robust scores for numerator and denominator  
    HZ = Z / zhat - (1 - Z) / (1 - zhat)  
    dr_num = yhat1 - yhat0 + (y - yhat) * HZ  
    dr_den = dhat1 - dhat0 + (D - dhat) * HZ  
    # Point estimate  
    point_est = np.mean(dr_num) / np.mean(dr_den)  
    # Standard error  
    psi = dr_num - point_est * dr_den  
    J_hat = np.mean(dr_den)  
    stderr = np.sqrt(np.mean(psi**2) / J_hat**2 / X.shape[0])  
    return point_est, stderr
```

Part IV: IV with Continuous Treatments

From LATE to DML-IV

Motivating Example: Demand Estimation

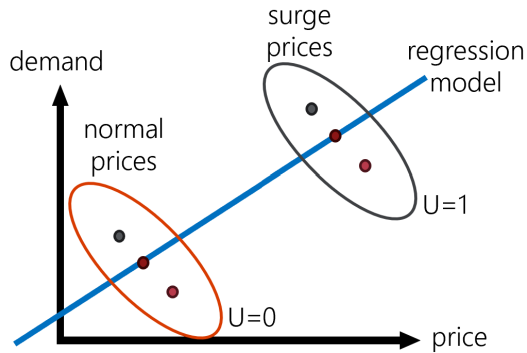
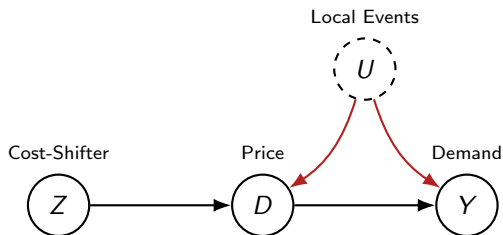
Suppose we want to estimate the causal effect of price (D) on product demand (Y).

- ▶ This is a **continuous** treatment.
- ▶ Price is almost always confounded. For example, high demand for a product often leads firms to increase its price.

The Need for an Instrument

We need an instrument Z that shifts the price but does not directly affect demand. A classic example is a **cost shifter**: a factor that changes the cost of producing the product (e.g., weather affecting crop yields, changes in raw material prices).

IV Regression Visually



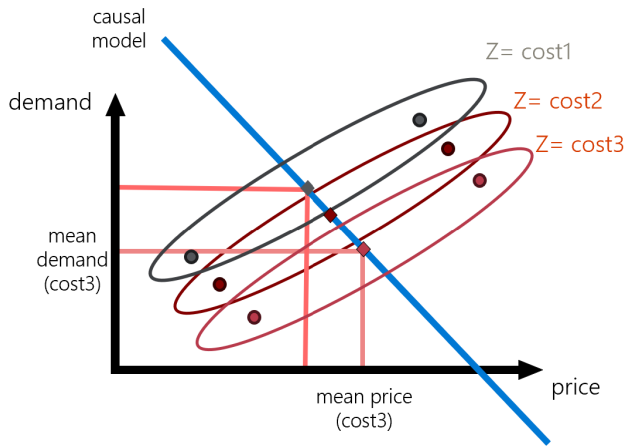
IV Regression Visually

Consider a simple linear model

$$Y = \theta D + U, \quad \mathbb{E}[U] = 0$$

Since U is uncorrelated with Z :

$$\begin{aligned}\mathbb{E}[Y | Z] &= \theta \mathbb{E}[D | Z] + \mathbb{E}[U | Z] \\ &= \theta \mathbb{E}[D | Z] + \mathbb{E}[U] \\ &= \theta \mathbb{E}[D | Z]\end{aligned}$$



The IV Formula for Continuous Treatments

Let's assume a simple linear structural model:

$$Y = \theta D + U$$

where U is an unobserved confounder, so $\text{Cov}(D, U) \neq 0$. We have an instrument Z with $\text{Cov}(Z, U) = 0$.

Taking the covariance of the equation with Z :

$$\text{Cov}(Y, Z) = \text{Cov}(\theta D + U, Z) = \theta \text{Cov}(D, Z) + \text{Cov}(U, Z)$$

Since $\text{Cov}(U, Z) = 0$, we can solve for θ :

The IV Estimator

$$\theta = \frac{\text{Cov}(Y, Z)}{\text{Cov}(D, Z)}$$

This is the famous Wald estimator, the ratio of the reduced form to the first stage.

The Analogy to OLS Normal Equations

Another way to think about IV is through moment conditions.

- ▶ OLS solves for the θ that makes the residual orthogonal to the treatment:

$$\mathbb{E}[(Y - \theta D) \cdot D] = 0$$

This fails when D is correlated with the residual (which contains U).

- ▶ IV solves for the θ that makes the residual orthogonal to the **instrument**:

$$\mathbb{E}[(Y - \theta D) \cdot Z] = 0$$

This works because the residual is uncorrelated with Z .

Double/Debiased ML for IV (DML-IV)

Now let's add high-dimensional controls X and allow for non-linearities.

Assume the outcome model is partially linear:

$$Y = \theta D + f(X) + U, \quad \mathbb{E}[U|X] = 0$$

And the instrument Z is conditionally exogenous given X : $\mathbb{E}[U|X, Z] = \mathbb{E}[U|X] = 0$.

Following the DoubleML logic, we can residualize out the effect of X :

$$\tilde{Y} = Y - \mathbb{E}[Y|X]$$

$$\tilde{D} = D - \mathbb{E}[D|X]$$

This leaves us with a simple residualized equation: $\tilde{Y} = \theta \tilde{D} + U$.

The DML-IV Algorithm

We can't use OLS on the residualized equation because $\tilde{D}$ is still correlated with U . Instead, we use the residualized instrument $\tilde{Z} = Z - \mathbb{E}[Z|X]$ and solve the moment equation:

$$\mathbb{E}[(\tilde{Y} - \theta\tilde{D})\tilde{Z}] = 0$$

The DML-IV Estimator

This gives the solution:

$$\hat{\theta} = \frac{\mathbb{E}_n[\check{Y}\check{Z}]}{\mathbb{E}_n[\check{D}\check{Z}]}$$

where $\check{Y}, \check{D}, \check{Z}$ are the cross-fitted residuals from regressing the variables on X .

Python Pseudocode for DML-IV

```
def dml_iv(X, Z, D, y, models, nfolds=5):
    # Cross-fit nuisance models  $E[V|X]$  for  $V$  in  $\{Y, D, Z\}$ 
    y_hat = cross_val_predict(models['y'], X, y, cv=nfolds)
    d_hat = cross_val_predict(models['d'], X, D, cv=nfolds)
    z_hat = cross_val_predict(models['z'], X, Z, cv=nfolds)
    # Compute residuals
    res_y = y - y_hat
    res_d = D - d_hat
    res_z = Z - z_hat
    # Point estimate
    point_est = np.mean(res_y * res_z) / np.mean(res_d * res_z)
    # Standard error
    epsilon = res_y - point_est * res_d
    var = np.mean(epsilon**2 * res_z**2) / np.mean(res_d * res_z)**2
    stderr = np.sqrt(var / X.shape[0])
    return point_est, stderr
```

Part V: DML Inference with Weak Instruments

What if the Instrument is Barely Relevant?

The Problem of Weak Instruments

What happens when the instrument Z is only weakly correlated with the treatment D ?

This means $\text{Cov}(D, Z)$ is close to zero.

Recall the IV estimator:

$$\hat{\theta} = \frac{\widehat{\text{Cov}}(Y, Z)}{\widehat{\text{Cov}}(D, Z)}$$

Unstable Estimator

If the denominator $\widehat{\text{Cov}}(D, Z)$ is close to zero, small amounts of sampling noise can cause huge swings in the estimate $\hat{\theta}$.

The distribution of $\hat{\theta}$ is no longer approximately normal, and our standard confidence intervals will be wrong.

Detecting Weak Instruments

Let the first-stage equation be $D = \pi_0 Z + \epsilon$, where Z is the (residualized) instrument and D is the (residualized) treatment.

The null hypothesis of a weak instrument test assumes π_0 is small, which induces a bias in the second-stage IV estimate $\hat{\theta}$.

- ▶ A frequently used heuristic for this is the **Nagar bias**, $\tau \approx 1/\lambda^2$, where $\lambda^2 = \pi_0^2/\sigma_\pi^2$ is a non-centrality parameter.
- ▶ We use a heteroskedastic-robust **F-statistic**: $F = \hat{\pi}^2/\hat{\sigma}_\pi^2$, where $\hat{\pi}$ is the OLS($D \sim Z$) estimate of π_0 and $\hat{\sigma}_\pi$ is the heteroskedasticity-robust s.e..
- ▶ Under the null that the Nagar bias is at most τ^* (e.g., 0.1), the F-statistic follows a non-central $\chi^2_{1,c}$ distribution with non-centrality parameter $c = \lambda^2 \leq 1/\tau^*$.
- ▶ With confidence α , we reject the null (conclude instrument is strong) if F is larger than $1 - \alpha$ percentile of $\chi^2_{1,1/\tau^*}$ distribution. *A common rule of thumb is $F > 10$.*

Robust Inference: The Anderson-Rubin (AR) Statistic

If we suspect a weak instrument, we should not use the standard confidence intervals. Instead, we can build a confidence set using a method that is robust to weak instruments.

Key Idea: Instead of testing a value of $\hat{\theta}$, we test a hypothesized value θ_0 .

Under the null hypothesis that the true effect is θ_0 , the residual $(Y - \theta_0 D)$ should be uncorrelated with Z . We can form a statistic based on this moment condition.

$$\mathbb{E}[(Y - \theta_0 D)Z] = 0$$

This moment is valid even if $\text{Cov}(D, Z)$ is zero!

The $C(\theta)$ Statistic

In the DML-IV setting, we use the cross-fitted residuals to form the score statistic $C(\theta)$.

The DML Weak-IV Robust Statistic

For a hypothesized value θ_0 , we compute:

- ▶ The moment function: $\check{M}(\theta_0) = \mathbb{E}_n[(\check{Y} - \theta_0 \check{D})\check{Z}]$
- ▶ The empirical variance: $\check{\Omega}(\theta_0) = V_n[(\check{Y} - \theta_0 \check{D})\check{Z}]$
- ▶ The score statistic: $C(\theta_0) = n \cdot \check{M}(\theta_0)' \check{\Omega}(\theta_0)^{-1} \check{M}(\theta_0)$
(for scalar Z : $C(\theta_0) = n \check{M}(\theta_0)^2 / \check{\Omega}(\theta_0)$)

Under the null $H_0 : \theta = \theta_0$, this statistic has a known distribution:

$$C(\theta_0) \xrightarrow{d} \chi^2(m)$$

where $m = \dim(Z)$.

Confidence Set by Inverting the Test

We can construct a 95% confidence set for θ by collecting all values θ_0 that we *fail to reject* at the 5% level.

Robust Confidence Set

$$CR_{95\%} = \{\theta_0 \in \Theta \mid C(\theta_0) \leq c_{0.95}\}$$

where $c_{0.95}$ is the 95th percentile of the $\chi^2(m)$ distribution.

This involves testing a grid of possible values for θ_0 and including those that yield a small $C(\theta_0)$ statistic.

The resulting set is a valid confidence set, even if the instrument is very weak. It might be a disjoint set or even the whole real line!

Python Pseudocode for Weak-IV Robust Inference

```
def weak_iv_confidence_set(check_y, check_d, check_z, grid, alpha=0.05):  
    n = check_y.shape[0]  
    crit_val = scipy.stats.chi2.ppf(1 - alpha, df=1) # m=1 instrument  
    accept_region = []  
    for theta_0 in grid:  
        moment = (check_y - theta_0 * check_d) * check_z  
        c_theta = n * np.mean(moment)**2 / np.var(moment)  
        if c_theta <= crit_val:  
            accept_region.append(theta_0)  
    return accept_region
```

Part VI: In-Class Activity & Takeaways

Putting IV into Practice

Activity: Spot the Instrument

Health Insurance and Emergency Room Visits

A state government wants to know whether expanding Medicaid coverage reduces emergency room (ER) visits. People who sign up for Medicaid may be systematically sicker or poorer than those who do not. In Oregon in 2008, the state expanded Medicaid through a lottery: eligible residents could enter a lottery, and winners were given the opportunity to apply for Medicaid.

Questions for the group:

- ▶ What is the instrument (Z), treatment (D), and outcome (Y)?
- ▶ State the exclusion restriction. Is it plausible?
- ▶ Who are the compliers?

Activity: Spot the Instrument

Online Recommendations and User Engagement

A travel website wants to estimate the causal effect of users becoming premium members on the number of days they visit the site. Users who choose to become premium members are likely already more engaged. The company runs an A/B test: a random subset of users sees an "easier signup" flow (fewer clicks, prominent button), while the control group sees the standard flow.

Questions for the group:

- ▶ What is the instrument (Z), treatment (D), and outcome (Y)?
- ▶ State the exclusion restriction. Is it plausible?
- ▶ Who are the compliers?

Takeaways

1. **IV for Unobserved Confounding:** Instrumental Variables are a powerful strategy for identification when conditional exogeneity fails due to unobserved confounders.
2. **LATE for Compliers:** For binary treatments, IV identifies the Local Average Treatment Effect (LATE)—the ATE for the subpopulation of *compliers* whose treatment status is changed by the instrument.
3. **Assumptions Matter:** The LATE interpretation relies on the exclusion restriction and monotonicity (no defiers). The IV strategy for continuous treatments relies on a partial linearity assumption.
4. **DR and DML for IV:** We can use our familiar Doubly Robust (for LATE) and Double ML (for continuous IV) machinery to get robust and efficient estimates.
5. **Weak Instruments Require Robust Inference:** When the first stage is weak, standard IV confidence intervals are unreliable. We must use methods like inverting the Anderson-Rubin statistic to get valid confidence sets.

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